

# NUDT2 (P50583): Functional Annotation Report

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**Gene:** NUDT2 (synonym APAH1) · **UniProt:** P50583 · **Organism:** *Homo sapiens* · **EC:** 3.6.1.17 · **Family:** Nudix hydrolase

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## Summary

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Human **NUDT2** (Nudix motif 2; also called APAH1) encodes a small (~17 kDa), soluble, intracellular **Nudix-family hydrolase** whose primary, defining biochemical activity is the **asymmetrical hydrolysis of diadenosine 5',5'''-P<sup>1</sup>,P<sup>4</sup>-tetraphosphate (Ap4A) into ATP + AMP** (EC 3.6.1.17). The term "asymmetrical" distinguishes NUDT2 from symmetrical Ap4A hydrolases (EC 3.6.1.41) that instead produce two molecules of ADP: NUDT2 cleaves the polyphosphate chain at the fourth phosphate counting from one of the bound adenosine moieties, so one adenosine departs carrying three phosphates (ATP) and the other carrying one (AMP). The enzyme adopts the canonical Nudix  $\alpha/\beta/\alpha$ -sandwich fold, uses a conserved Nudix-box active site, requires a divalent metal cofactor ( $Mg^{2+}$  or  $Zn^{2+}$ ), and acts on diadenosine polyphosphates bearing four or more phosphates (Ap<sub>n</sub>A,  $n \geq 4$ ) as its preferred substrate class.

Functionally, NUDT2 is best understood as the **catabolic "off-switch" of the Ap4A second-messenger / stress-alarmon system**, a role that is conserved from bacteria to humans. Ap4A is synthesized largely as a side reaction of aminoacyl-tRNA synthetases—most notably lysyl-tRNA synthetase (LysRS)—and accumulates under immunological activation and genotoxic/metabolic stress. By degrading Ap4A, NUDT2 both terminates the signal and regenerates purinergically active mononucleotides. Through this activity NUDT2 tunes at least three downstream programs: (1) the **LysRS → Ap4A → Hint1 → MITF/USF2 transcriptional axis** that drives gene expression in activated mast cells and cardiomyocytes; (2) a **DNA-damage replication checkpoint**, in which Ap4A accumulates after DNA damage and inhibits the *initiation* of DNA replication; and (3) **interferon/immune and tumor-promotion gene programs**, which shift markedly when intracellular Ap4A is raised by NUDT2 disruption.

A second, physiologically critical activity has emerged more recently: NUDT2 is also a **bifunctional m<sup>7</sup>G mRNA 5'-decapping enzyme**. Biallelic loss-of-function variants in NUDT2 cause a **recessive neurodevelopmental disorder**, and add-back genetics demonstrate that it is the loss of the *decapping* activity (not Ap4A hydrolysis) that disrupts mRNA homeostasis and drives the molecular phenotype. NUDT2 therefore sits at the intersection of nucleotide-signaling catabolism and RNA-cap metabolism, operating in both the **cytoplasm and the nucleus** (its *Drosophila* ortholog is predominantly nuclear, associating with euchromatin/facultative heterochromatin). The sections below detail the evidence for each of these roles and the mechanistic model that unifies them.

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## Key Findings

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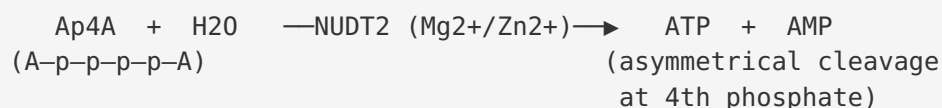
### Finding 1 — NUDT2 is an asymmetrical Ap4A hydrolase (EC 3.6.1.17) that cleaves Ap4A into ATP + AMP

The core, defining function of NUDT2 is enzymatic: it is an **asymmetrical diadenosine tetraphosphate hydrolase**. As a Nudix (nucleoside diphosphate linked to X) hydrolase, it cleaves the dinucleoside polyphosphate Ap4A at the fourth phosphate counting from one bound adenosine, generating **ATP + AMP**. This asymmetrical cleavage pattern is the biochemical signature that separates NUDT2 (EC 3.6.1.17) from symmetrical bacterial-type enzymes (e.g., ApaH, EC 3.6.1.41) that yield 2 × ADP.

The crystal structure of the human enzyme confirms a canonical Nudix architecture. As reported in the structural study of wild-type and mutant human Ap4A hydrolase, "*Ap4A hydrolase (asymmetrical diadenosine tetraphosphate hydrolase, EC 3.6.1.17), an enzyme involved in a number of biological processes, is characterized as cleaving the polyphosphate chain at the fourth phosphate from the bound adenosine moiety,*" and "*Similar to the canonical Nudix fold, human Ap4A hydrolase shows the common αβ-sandwich architecture*" (PMID: [23384440](#)). Conserved active-site residues (including E58) coordinate the substrate phosphates and the catalytic divalent metal; catalysis is Mg<sup>2+</sup>-dependent and can be inhibited by fluoride via formation of an MgF<sub>3</sub><sup>-</sup> transition-state mimic.

The orthologous enzyme from *Drosophila melanogaster* provides quantitative kinetic confirmation that "*diadenosine tetraphosphate is hydrolysed to ATP and AMP*," with  $K_m \approx 9\text{--}12\ \mu\text{M}$  and  $k_{cat} \approx 13\text{--}43\ \text{s}^{-1}$  and a requirement for a divalent cation ( $\text{Mg}^{2+}$  or  $\text{Zn}^{2+}$ ) (PMID: 17344088). Kinetic parameters for the human enzyme have also been measured directly (PMID: 12370170).

#### Reaction catalyzed:



### Finding 2 — NUDT2 controls the LysRS–Ap4A–Hint1–MITF transcriptional axis as the Ap4A-degrading "off-switch"

Beyond simple metabolite clearance, NUDT2 is an active regulator of a defined signaling cascade. In immunologically activated (FcεRI-stimulated) mast cells, LysRS is phosphorylated on Ser207 in a MAPK-dependent manner, released from the multi-tRNA-synthetase complex, and translocated to the nucleus, where it synthesizes the second messenger **Ap4A**. Ap4A then binds the histidine-triad protein **Hint1**, releasing the transcription factors **MITF** and **USF2** from Hint1-mediated repression and enabling transcription of their target genes (PMID: 14975237; PMID: 19524539; PMID: 23159739).

NUDT2 is the enzyme that terminates this signal. The study of the transcriptional regulation network in activated mast cells provides direct evidence: "*we provided here evidence that the 'Nudix' type 2 gene product, Ap(4)A hydrolase, is responsible for Ap(4)A degradation following the immunological activation of mast cells. The knockdown of Ap(4)A hydrolase modulated Ap(4)A accumulation, resulting in changes in the expression of MITF and USF2 target genes*" (PMID: 18644867). The same axis operates in cardiac cells activated with the  $\beta$ -agonist isoproterenol. The establishment of a dedicated metabolizing enzyme (NUDT2) satisfied a key criterion for defining Ap4A as a bona fide second messenger. As summarized for the signaling pathway, Ap4A "*is accumulated intracellularly above 700 microM in IgE-Ag-activated mast cells, binds to Hint, liberates MITF, and thus leads to the activation of MITF-dependent gene expression*" (PMID: 14975237), and the structural switch of LysRS "*triggers LysRS-directed production of the second messenger Ap(4)A that activates MITF*" (PMID: 23159739).

### Finding 3 — NUDT2 is a bifunctional mRNA decapping enzyme; loss causes a recessive neurodevelopmental disorder

A landmark clinical-genetic study revealed a second, distinct catalytic activity of NUDT2 and tied it to human disease. Biallelic NUDT2 variants (novel missense and in-frame deletions) were identified in **18 children/young adults from 10 families** presenting with a recessive neurodevelopmental disorder: intellectual disability, motor developmental delay, gait disturbance, sometimes peripheral neuropathy, and neuroimaging findings including corpus callosum abnormalities and basal ganglia deposits. According to the study, *"The disorder is associated with rare variants in NUDT2, a mRNA decapping and Ap4A hydrolysing enzyme, including novel missense and in-frame deletion variants. We show that these NUDT2 variants lead to a marked loss of enzymatic activity"* (PMID: 38141063).

Crucially, this establishes NUDT2 as a **bifunctional protein**—both an mRNA-decapping and an Ap4A-hydrolysing enzyme. NUDT2-deficient patient fibroblasts show an altered transcriptome with changes in mRNA half-life/stability, including up-regulation of interferon-responsive/host-response genes. To disentangle which activity is responsible, the study used add-back genetics: *"add-back experiments using an Ap4A hydrolase defective in mRNA decapping highlighted loss of NUDT2 decapping as the activity implicated in altered mRNA homeostasis"* (PMID: 38141063). This elegantly demonstrates that the **decapping activity**, not Ap4A hydrolysis, drives the mRNA phenotype that underlies the disease. The decapping activity is mechanistically consistent with the broader Nudix family: these enzymes act on m<sup>7</sup>G/RNA 5' caps (PMID: 32059949), and bacterial Ap4A hydrolases efficiently remove pyrophosphate from 5' mRNA termini (PMID: 23184251).

### Finding 4 — NUDT2 sets intracellular Ap4A levels that act as a stress/DNA-damage alarmone and modulate immune/cancer gene programs

Because NUDT2 is the principal Ap4A-degrading enzyme in human cells, it is the main determinant of intracellular Ap4A concentration, and Ap4A in turn functions as a stress/DNA-damage signal. Ap4A rises several-fold after DNA interstrand-crosslinking damage (e.g., mitomycin C) and in DNA-repair mutants (XRCC1, PARP1, APTX, FANCG). DNA ligase III synthesizes damage-induced Ap4A, which then inhibits the **initiation** (but not elongation) of DNA replication—*"Ap4A was found to cause a marked inhibition of the initiation of DNA replicons, while elongation was unaffected"*—with 70–80% inhibition at 20 μM, supporting a model in which Ap4A is an inducible ligand in the DNA-damage response that prevents replication of damaged DNA (PMID: 26204256).

The magnitude of NUDT2's control over Ap4A is striking. A comparative RNA-Seq analysis compared KBM-7 chronic myelogenous leukemia cells with KBM-7 cells *"in which the NUDT2 Ap4A hydrolase gene had been disrupted (NuKO cells), causing a 175-fold increase in intracellular Ap4A"* (PMID: 27144453). This increase differentially expressed 6,288 genes ( $P < 0.05$ ): down-regulated gene sets were enriched for interferon responses, pattern-recognition receptors, inflammation, and tumor-promotion categories (EMT, proliferation, invasion, metastasis), with predicted upstream regulators NF- $\kappa$ B, STAT1/2, IRF3/4, and SP1; some pro-apoptotic genes were up-regulated. These data position NUDT2 as a modulator of immune and cancer-related transcriptional programs via its control of Ap4A.

### Finding 5 — NUDT2 is an intracellular enzyme acting in the nucleus and cytoplasm; broad Ap<sub>n</sub>A ( $n \geq 4$ ) substrate range with divalent-cation dependence

NUDT2 is a small (~17 kDa), soluble, intracellular Nudix hydrolase lacking a secretion signal. Subcellular localization evidence from the *Drosophila* ortholog (Apf, 16.6 kDa) indicates a nuclear site of action: *"Subcellular localization with Apf-EGFP fusion constructs reveals Apf to be predominantly nuclear, having an apparent preferential association with euchromatin and facultative heterochromatin. This supports a nuclear function for diadenosine tetraphosphate"* (PMID: 17344088). The enzyme *"always produces an NTP product, with substrate preference depending on pH and divalent ion (Zn(2+) or Mg(2+))"*, hydrolyzing  $\text{Ap4A} \rightarrow \text{ATP} + \text{AMP}$  and longer diadenosine polyphosphates (e.g.,  $\text{Ap6A} \rightarrow \text{ATP}$ ) (PMID: 17344088).

In human cells, NUDT2 function is manifested in both compartments: in the **nucleus**, degrading Ap4A that regulates MITF/USF2 transcription (PMID: 18644867); and in **cytoplasmic/nuclear mRNA turnover**, via its decapping activity affecting transcriptome stability (PMID: 38141063). Related and bacterial asymmetrical Ap4A hydrolases also prefer Ap<sub>n</sub>A with  $\geq 4$  phosphates and can act on 5' mRNA termini (PMID: 23184251). Notably, this subfamily of Nudix enzymes (those active on dinucleoside polyphosphates with  $\geq 4$  phosphates) additionally exhibits **PRPP pyrophosphatase** activity in vitro, generating ribose 1,5-bisphosphate—though for the human enzyme the specificity constant for PRPP is low and its physiological relevance is uncertain (PMID: 12370170).

## Finding 6 — NUDT2 is the catabolic/signal-terminating enzyme of the evolutionarily conserved Ap4A alarmone system (bacteria → humans)

Ap4A and related dinucleoside polyphosphates are ubiquitous from bacteria to humans and are increasingly viewed as conserved stress "alarmones"/second messengers. A recent review frames *"the notion that AP4A is a conserved second messenger in organisms ranging from bacteria to humans and is able to signal and modulate cellular stress regulation"* as a promising unifying concept (PMID: 37223742). These dinucleotides are synthesized largely as a side reaction of aminoacyl-tRNA synthetases (notably LysRS) from aminoacyl-AMP + ATP: *"most aaRSs can also produce dinucleotide polyphosphates in a variety of physiological conditions. The dinucleotide polyphosphates produced by aaRS are biologically active both extra- and intra-cellularly, and seem to function as important signaling molecules"* (PMID: 23536246; see also PMID: 14975237).

Degradation is the essential counterpart to synthesis. As articulated for adenine dinucleotide signaling, *"The enzymatic cleavage of the dinucleotides plays a dual role for their biological function: (a) termination of the signal; and (b) generation of purinergically active products such as ATP, ADP and finally adenosine"* (PMID: 9131408). NUDT2 is the dedicated mammalian asymmetrical Ap4A hydrolase performing this catabolic step (PMID: 18644867; PMID: 27144453). Mechanistically it belongs to the Nudix "housecleaning" hydrolase family (PMID: 8810257), which hydrolyzes nucleoside-diphosphate-X substrates using a conserved Nudix box and divalent-metal catalysis (PMID: 17344088; PMID: 23384440).

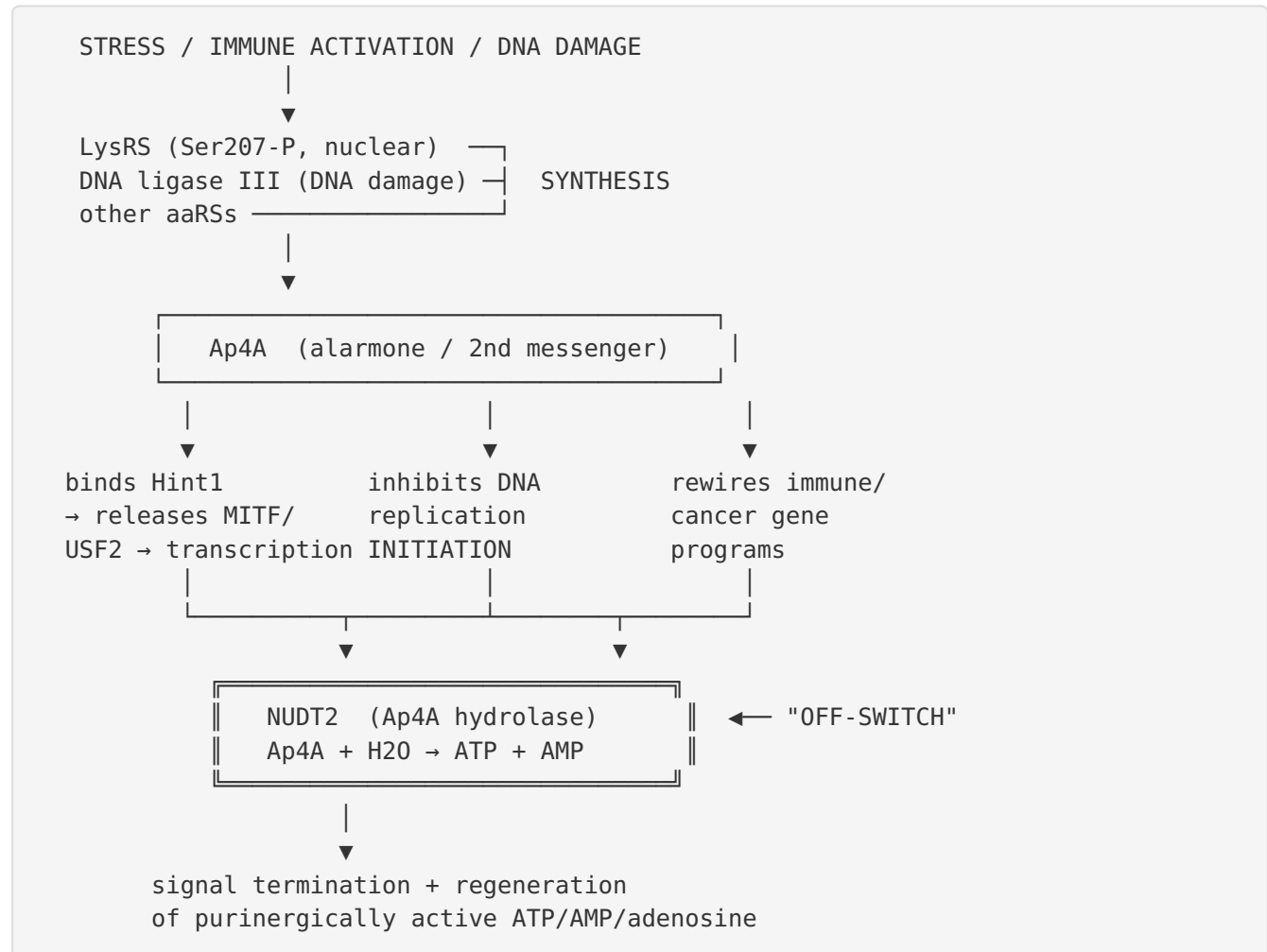
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## Mechanistic Model / Interpretation

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NUDT2 can be understood as a **dual-function hydrolase** sitting at the hub of two nucleotide-metabolism networks: (1) the **Ap4A alarmone cycle**, where it is the signal-terminating catabolic enzyme; and (2) **mRNA cap metabolism**, where it acts as a decapping enzyme governing transcript stability. Both activities derive from the same Nudix  $\alpha/\beta/\alpha$ -sandwich fold and conserved metal-dependent active site, consistent with the well-documented substrate ambiguity of Nudix hydrolases.

## The Ap4A signaling cycle



In this model, Ap4A levels represent a dynamic balance between synthesis (aaRSs, DNA ligase III) and degradation (NUDT2). When NUDT2 activity is reduced—whether by knockdown, genetic knockout, or disease-causing variants—Ap4A rises dramatically (up to 175-fold in NuKO cells), amplifying and prolonging all downstream Ap4A-dependent effects: enhanced MITF/USF2 transcription, suppression of DNA replication initiation, and remodeling of interferon/immune and tumor-promotion gene networks. Conversely, robust NUDT2 activity keeps Ap4A low, maintaining the system poised to respond to the next stress stimulus.

## The decapping / mRNA-homeostasis function

m7G-capped mRNA —NUDT2 decapping→ altered 5' end  
→ changed mRNA  
half-life/stability

The disease genetics make clear that these two activities are **separable**: an engineered NUDT2 that retains Ap4A hydrolysis but is defective in decapping fails to rescue mRNA homeostasis, whereas the clinical phenotype tracks specifically with decapping loss. This argues that, at least for the neurodevelopmental disorder, NUDT2's RNA-cap activity is the physiologically limiting function in neurons, even though its Ap4A-hydrolase role dominates the historical literature.

### Comparative substrate / mechanism table

| Property            | NUDT2 (human, asymmetrical)                          | Symmetrical ApaH-type | Ecto-PDE I (contrast)          |
|---------------------|------------------------------------------------------|-----------------------|--------------------------------|
| EC number           | 3.6.1.17                                             | 3.6.1.41              | 3.1.4.1                        |
| Ap4A products       | <b>ATP + AMP</b>                                     | 2 × ADP               | NMP + Ap <sub>n-1</sub> (ecto) |
| Fold / family       | Nudix (α/β/α)                                        | —                     | alkaline phosphodiesterase     |
| Metal cofactor      | Mg <sup>2+</sup> / Zn <sup>2+</sup>                  | —                     | —                              |
| Location            | <b>Intracellular</b> (nucleus + cytoplasm)           | intracellular         | <b>cell-surface / ecto</b>     |
| Preferred substrate | Ap <sub>n</sub> A, n ≥ 4; also m <sup>7</sup> G caps | Ap <sub>n</sub> A     | broad di-/mononucleotides      |

It is important not to conflate NUDT2 with the **ecto** diadenosine-polyphosphate hydrolases found at cell surfaces (e.g., PC-1/alkaline phosphodiesterase-I activity in airway epithelia and adrenal chromaffin membranes; [PMID: 10919994](#), [PMID: 9784621](#)). Those enzymes are a different family (PDE-I/ENPP), are membrane-bound/extracellular, and metabolize extracellular dinucleotides that act on purinergic receptors—distinct from NUDT2's intracellular role.

## Evidence Base

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| PMID                     | Title (abbrev.)                                                           | How it supports/challenges the findings                                                                                                                                              |
|--------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">23384440</a> | <i>Crystal structure of human Ap4A hydrolase</i>                          | Defines EC 3.6.1.17, asymmetrical cleavage at 4th phosphate, and Nudix $\alpha\beta\alpha$ fold of the human enzyme (F1).                                                            |
| <a href="#">17344088</a> | <i>Bis(5'-nucleosyl)-tetraphosphatase from Drosophila</i>                 | Kinetics (Ap4A $\rightarrow$ ATP+AMP), Ap <sub>n</sub> A (n $\geq$ 4) substrate range, Mg <sup>2+</sup> /Zn <sup>2+</sup> dependence, nuclear localization of ortholog (F1, F5, F6). |
| <a href="#">18644867</a> | <i>Ap4A hydrolase in mast-cell transcription</i>                          | Directly identifies NUDT2 as the Ap4A-degrading enzyme regulating MITF/USF2 target genes (F2, F5).                                                                                   |
| <a href="#">14975237</a> | <i>LysRS and Ap4A regulate MITF</i>                                       | Establishes the LysRS $\rightarrow$ Ap4A $\rightarrow$ Hint1 $\rightarrow$ MITF axis that NUDT2 terminates (F2, F6).                                                                 |
| <a href="#">19524539</a> | <i>LysRS signaling in immune response</i>                                 | Ser207 phosphorylation of LysRS governs Ap4A production in vivo (F2).                                                                                                                |
| <a href="#">23159739</a> | <i>Structural switch of LysRS</i>                                         | Ap4A as the LysRS-produced second messenger activating MITF (F2).                                                                                                                    |
| <a href="#">38141063</a> | <i>Biallelic NUDT2 variants cause neurodevelopmental disease</i>          | Establishes bifunctional decapping + Ap4A-hydrolase activity; decapping loss drives mRNA phenotype and disease (F3).                                                                 |
| <a href="#">32059949</a> | <i>Nucleic-acid binding of human Nudix enzymes</i>                        | Supports Nudix action on m <sup>7</sup> G/RNA caps, consistent with NUDT2 decapping (F3).                                                                                            |
| <a href="#">23184251</a> | <i>Substrate ambiguity among Nudix hydrolases</i>                         | Bacterial Ap4A hydrolases remove 5' mRNA pyrophosphate; rationalizes dual activity and Ap <sub>n</sub> A preference (F3, F5).                                                        |
| <a href="#">26204256</a> | <i>Ap4A synthesized on DNA damage inhibits replication initiation</i>     | Defines the DNA-damage alarmone role of NUDT2's substrate (F4).                                                                                                                      |
| <a href="#">27144453</a> | <i>NUDT2 disruption elevates Ap4A, down-regulates immune/cancer genes</i> | NUDT2 as principal determinant of intracellular Ap4A (175-fold); links to immune/cancer programs (F4).                                                                               |
| <a href="#">37223742</a> | <i>The mysterious Ap4A</i>                                                | Frames Ap4A as a conserved bacteria-to-human second messenger (F6).                                                                                                                  |

| PMID                     | Title (abbrev.)                                      | How it supports/challenges the findings                                                                        |
|--------------------------|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <a href="#">23536246</a> | <i>aaRS generate dinucleotide polyphosphates</i>     | Identifies aaRSs as the synthetic source of Ap4A (F6).                                                         |
| <a href="#">9131408</a>  | <i>Adenine dinucleotides as signalling molecules</i> | Hydrolysis both terminates signal and yields active products (F6).                                             |
| <a href="#">8810257</a>  | <i>MutT/Nudix "housecleaning" enzymes</i>            | Family context: NUDT2 is a Nudix housecleaning hydrolase (F6).                                                 |
| <a href="#">12370170</a> | <i>Nudix PRPP pyrophosphatase activity</i>           | Human Ap4A hydrolase kinetics; secondary PRPP activity of the Ap <sub>n</sub> A-preferring subfamily (F1, F5). |
| <a href="#">40807231</a> | <i>Ap4A in cancer</i>                                | Reviews Ap4A biosynthesis/degradation, targets, and cancer relevance (F4, F6).                                 |

**Contrasting / cautionary evidence.** The ecto-enzyme studies ([PMID: 10919994](#), [PMID: 9784621](#)) show that extracellular Ap4A hydrolysis in human tissues is performed by membrane-bound PDE-I/ENPP enzymes—**not** NUDT2—underscoring that NUDT2's functional niche is strictly intracellular. The Nudix substrate-ambiguity review ([PMID: 23184251](#)) cautions that in vitro multispecificity (e.g., PRPP, oxidized nucleotides) can lead to annotation errors, so secondary activities should be interpreted with care.

## Limitations and Knowledge Gaps

- 1. Structure-based mechanistic detail of human decapping is limited.** The crystal structure defines the Ap4A-hydrolase active site ([PMID: 23384440](#)), but how the same active site engages an m<sup>7</sup>G cap, and the precise residues that distinguish decapping from Ap4A hydrolysis, are inferred largely from the "decapping-defective" add-back construct ([PMID: 38141063](#)) rather than from a cap-bound structure.
- 2. Direct human (vs. ortholog) localization data are sparse.** The strongest subcellular-localization evidence (predominantly nuclear, euchromatin-associated) comes from the *Drosophila* Apf-EGFP fusion ([PMID: 17344088](#)); rigorous endogenous localization of human NUDT2 across cell types and stress states is less well established.

3. **Causation vs. correlation in gene-program changes.** The large transcriptomic shifts upon NUDT2 disruption ([PMID: 27144453](#); [PMID: 38141063](#)) are high-throughput and associative. Which changes are driven by elevated Ap4A versus by loss of decapping (and which are direct vs. secondary) is only partially resolved.
  4. **Physiological relevance of secondary activities.** In vitro PRPP pyrophosphatase activity ([PMID: 12370170](#)) has a low specificity constant for the human enzyme and no established in vivo role; the "housecleaning" antimutator activities noted for Nudix enzymes ([PMID: 8810257](#); [PMID: 23184251](#)) are likely residual for NUDT2.
  5. **Second-messenger status of Ap4A remains debated.** While the LysRS–Ap4A–Hint1–MITF axis is well supported, the broader claim that Ap4A is a universal second messenger is still under active re-evaluation in the field, and quantitative in-cell Ap4A concentrations vary widely between studies.
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## Proposed Follow-up Experiments / Actions

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1. **Co-crystallize/cryo-EM human NUDT2 with an m<sup>7</sup>G-cap analogue** (and with a non-hydrolyzable Ap4A analogue such as AppCH<sub>2</sub>ppA) to map the structural basis for its dual substrate recognition and to identify residues that selectively support decapping vs. Ap4A hydrolysis—directly extending the decapping-defective mutant logic of [PMID: 38141063](#).
2. **Separation-of-function rescue in neuronal models.** Express decapping-only and Ap4A-hydrolase-only NUDT2 variants in patient-derived iPSC neurons/organoids to determine which activity rescues the neurodevelopmental mRNA-homeostasis phenotype, and to define affected transcripts (5'-end sequencing / cap-status profiling).
3. **Endogenous human localization and dynamics.** Use CRISPR knock-in of a fluorescent/epitope tag to quantify NUDT2's nuclear vs. cytoplasmic distribution in human cells at rest and after immune activation or DNA damage, testing whether it is recruited to sites of Ap4A synthesis.
4. **Quantitative Ap4A flux measurements.** Apply targeted LC-MS/MS to measure intracellular Ap4A synthesis and degradation rates as a function of NUDT2 activity across stress conditions, resolving the synthesis/degradation balance that sets alarmone levels.

5. **Dissect downstream causality.** Combine NUDT2 knockout with Ap4A-insensitive Hint1/MITF mutants (or LysRS-Ser207 mutants) to determine how much of the immune/cancer transcriptional rewiring ([PMID: 27144453](#)) is Ap4A-dependent versus decapping-dependent.
  6. **Therapeutic exploration.** Given Ap4A's roles in cancer adaptation ([PMID: 40807231](#)) and the availability of selective asymmetrical-Ap4A-hydrolase inhibitors (e.g., phosphorothioylated polyols,  $K_i \approx 1.5 \mu\text{M}$  for the human enzyme; [PMID: 12697025](#)), evaluate NUDT2 modulation as a tool to tune Ap4A for anti-tumor or DNA-damage-sensitization strategies.
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## Gene Identity Verification

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The research target was confirmed throughout: gene symbol **NUDT2** (synonym APAH1), UniProt **P50583**, *Homo sapiens*, encoding **bis(5'-nucleosyl)-tetrphosphatase [asymmetrical] / Ap4A hydrolase (EC 3.6.1.17)**, a member of the **Nudix hydrolase family** (Nudix motif 2). All primary literature cited describes this specific enzyme or its direct orthologs; no ambiguity with unrelated same-symbol genes was encountered. Domain annotations (NUDIX\_hydrolase domain, Tetra\_PHTase) are fully consistent with the catalytic function documented in the literature.

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